Project 2: Advanced Data, APIs, & Sensor Integration

SDG Problem Explanation

Selected SDG:
SDG 4 - Quality Education (Focusing on effective learning and student engagement).

The Problem:
University students, particularly those in fast-paced or technical degree programs, frequently experience burnout and a lack of motivation when managing independent study hours and complex coursework. Traditional to-do lists and study tracking often feel tedious.

Why It Matters (Impact):
Inconsistent study habits lead to severe procrastination, heightened anxiety, and lower academic performance. Without a structured and engaging way to manage self-guided learning, students fail to build the consistent habits necessary for long-term career success.

The Solution (EduQuest):
EduQuest tackles this by transforming independent learning into a highly engaging, gamified experience. By utilising RPG mechanics such as creating custom "Side Quests" for daily tasks, answering interactive quizzes, and rewarding consistency with XP, dynamic titles, and daily streaks, the app encourages proactive time management and makes quality education a rewarding daily habit.

Project 2 Enhancements

Continuing the SDG 4 Mission:

Project 2 builds on EduQuest's original gamification system by connecting it to real-world data, persistent storage, and hardware sensors. The goal remains the same - helping students build consistent, motivated study habits - but now the app is fully connected, persistent, and aware of the student's physical environment.

Local Persistence (Room Database):

A student's profile, XP, streak, course progress, and check-in history are now saved permanently on the device using Room. This means progress is never lost when the app is closed, and the same device can support multiple students — each logging in with their own matric number to restore their personal progress.

Cloud Integration (Firebase Firestore):

A Campus Leaderboard powered by Firebase Firestore introduces a social, community-driven element to EduQuest. Whenever a student checks in on campus, their latest XP and title are synced to the cloud, allowing all students across different devices to see how they rank against their peers in real time — turning individual study habits into a shared, motivating experience.

Data from the Internet (Overpass REST API):

The new Study Map screen connects to the OpenStreetMap Overpass API via Retrofit to fetch real, live data about nearby libraries, cafes, universities, and parks. This directly supports the SDG 4 goal of making quality education accessible — students are guided towards real physical spaces conducive to studying, wherever they are.

Hardware Sensor Integration (GPS/Location):

A new Campus Check-In screen uses the device's GPS sensor through Android's Fused Location Provider to detect when a student is physically present on campus. Being on campus rewards the student with bonus XP and contributes to their daily streak — reinforcing the habit of showing up and engaging with their learning environment, both digitally and physically.